STM32F3 Technical Training

For reference only

Refer to the latest documents for details

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Direct memory access controller (DMA)

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DMA Features

- 12 independently configurable channels: hardware requests or software trigger on each channel.
    - DMA1: 7 Channels
    - DMA2: 5 Channels
- Software programmable priorities: Very high, High, Medium or Low. (Hardware priority in case of equality).
- Programmable and Independent source and destination transfer data size: Byte, Halfword or Word.
- 3 event flags for each channel: DMA Half Transfer, DMA Transfer complete and DMA Transfer Error.
- Memory-to-memory, peripheral-to-memory and memory-to-peripheral transfers and peripheral-to-peripheral transfers.
- Faulty channel is automatically hardware disabled in case of bus access error.
- Programmable number of data to be transferred: up to 65535.
- Support for circular buffer management.

DMA1 Request Mapping (1/2)

Periphera l	Channel 1	Channel 2	Channel 3	Channel 4	Channel 5	Channel 6	Channel 7
ADC	ADC1						
SPI		SPI1_RX	SPI1_TX	SPI2_RX	SPI2_TX		
USART		USART3_TX	USART3_RX	USART1_TX	USART1_RX	USART2_RX	USART2_TX
I2C				I2C2_TX	I2C2_RX	I2C1_TX	I2C1_RX
TIM1 (*)		TIM1_CH1	TIM1_CH2	TIM1_CH4 TIM1_TRIG TIM1_COM	TIM1_UP	TIM1_CH3	
TIM2	TIM2_CH3	TIM2_UP				TIM2_CH1	TIM2_CH2 TIM2_CH4
TIM3		TIM3_CH3	TIM3_CH4 TIM3_UP			TIM3_CH1 TIM3_TRIG	
TIM4	TIM4_CH1			TIM4_CH2	TIM4_CH3		TIM4_UP
TIM6 / DAC (*)			TIM6_UP DAC_CH1 (1)				

- (*) Available on STM32F30x only.
- (1) DMA request mapped on this DMA channel only if the corresponding remapping bit is set in the SYSCFG_CFGR1 register

DMA1 Request Mapping (2/2)

Peripherals	Channel 1	Channel 2	Channel 3	Channel 4	Channel 5	Channel 6	Channel 7
TIM7 / DAC (*)				TIM7_UP DAC_CH2 (1)			
TIM16			TIM16_CH1 TIM16_UP			TIM16_CH1 TIM16_UP (*) (1)	
TIM17	TIM17_CH 1 TIM17_UP						TIM17_CH1 TIM17_UP (*) (1)
TIM18 / DAC channel 3 (**)					TIM18_UP DAC_CH3		
TIM19 (**)	TIM19_CH3 TIM19_CH4	TIM19_CH1	TIM19_CH2	TIM19_UP			

- (*) Available on STM32F30x only
- (**) Available on STM32F37x only.
- (1) DMA request mapped on this DMA channel only if the corresponding remapping bit is set in the SYSCFG_CFGR1 register

DMA2 Request Mapping

Peripherals	Channel1	Channel2	Channel3	Channel4	Channel5
ADC	ADC2	ADC4	ADC2 (1) SDADC1	ADC4 (1) SDADC2	ADC3 SDADC3
SPI3	SPI3_RX	SPI3_TX			
UART4(*)			UART4_RX		UART4_TX
TIM6 / DAC channel 1			TIM6_UP DAC_CH1		
TIM7 / DAC channel 2				TIM7_UP DAC_CH2	
TIM8 / DAC (*)	TIM8_CH3 TIM8_UP	TIM8_CH4 TIM8_TRIG TIM8_COM	TIM8_CH1		TIM8_CH2
TIM18 / DAC channel 3 (**)					TIM18_UP DAC_CH3

- (*) Available on STM32F30x only
- (**) Available on STM32F37x only.

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Quiz

- How many DMA Channels are available in the STM32F3xx ?

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- How many interrupts can be generated for each channel?

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- Which Channel is able to perform Memory to Memory transfer?

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Thank you !

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